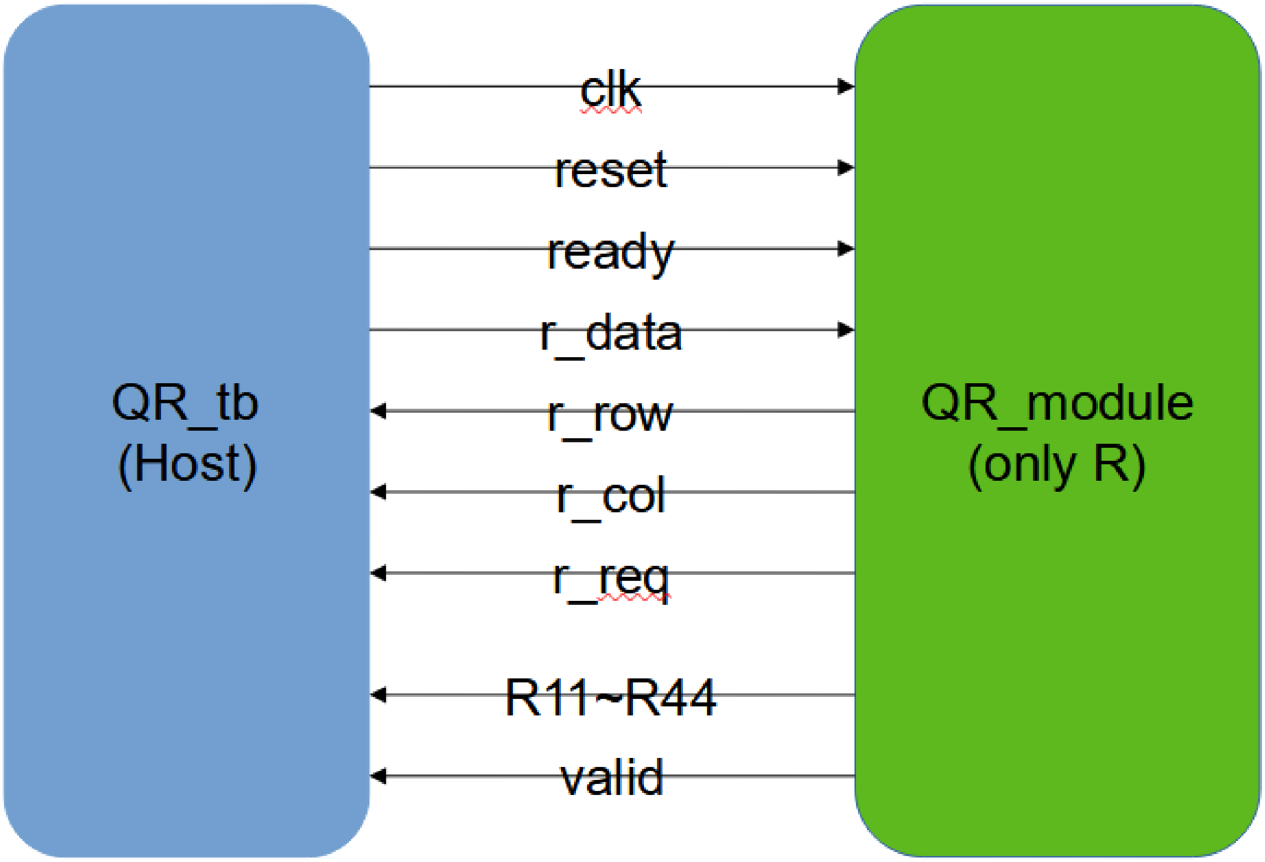
# VLSIDSP Final Project 電機四B 4111064113 白舜文

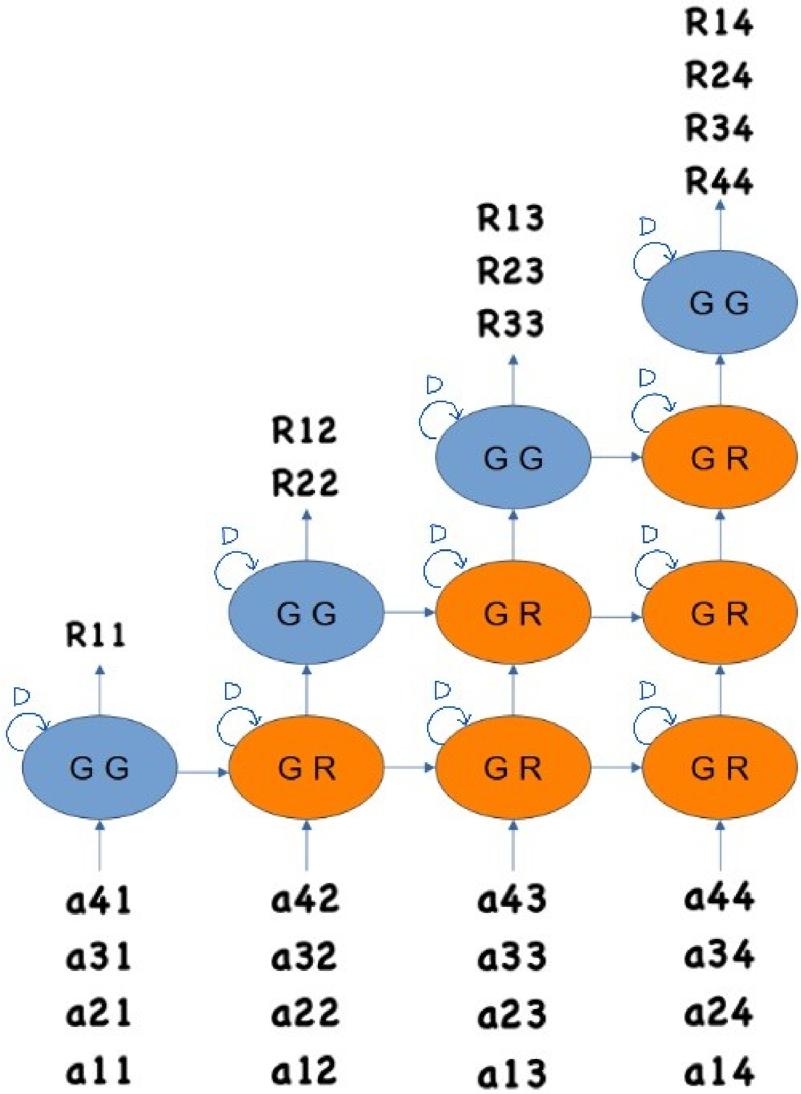
## I. Problem Description & Block Diagram Design

此次實作的 QR factorization module，將 4X4 的 A 矩陣經 QR 分解後得到 4X4 的上三角矩陣 R，並配合 Givens Generation(GG, rotation mode)與 Givens Rotation(GR, vectoring mode) PEs 做旋轉動作，GG 決定旋轉的方向，GR 接受 GG 的方向訊號後進行旋轉。主要的運算模組為 folded CORDIC processor，將硬體簡化至加法器與移位器，原本的 iteration number 為 12，採用 folded 方法後，one stage 內可一次計算四個 iteration，將 computing latency 降至 3 個 clock cycles，最後 scaling 得到最終的結果。以下為架構圖：




| Port | I/O | Width | Desciption | Port | I/O | Width | Desciption |
|---|---|---|---|---|---|---|---|
| clk | I | 1 | Postive trigger clock | r_row | O | 2 | A matrix row |
| reset | I | 1 | asynchrnous | r_col | O | 2 | A matrix column |
| ready | I | 1 | Active high 開始計算 | r_req | O | 1 | External RAM request |
| r_data | I | 8 | Read A matrix | valid | O | 1 | 完成計算 |
| R11~R44 | O | 12 | R element output | | | | |

而以下即為此次 QR 分解所需之 DFG，分別使用 4 個 GG 與 6 個 GR 實現：




## II. Matlab verification

此次使用 Matlab 進行簡單的 simulation，為 Q5.7 的 fixed-point format，採取 bottom-up 的方式對 A 矩陣進行 Givens Rotation 的整理運算，最終得到一個 4X4 的上三角矩陣 R，以下為模擬所得結果：

```
Command Window
=== 原始 A 矩陣 ===
     5     2     1     0
     2     5     2     1
     1     2     5     2
     0     1     2     5

=== Ideal R matrix ===
    5.4772    4.0166    2.5560    0.7303
         0    4.2269    3.2490    2.6182
         0         0    4.1122    2.8274
         0         0         0    3.8233

=== CORDIC R 矩陣 ===
    5.5000    4.0078    2.5391    0.7188
         0    4.2578    3.2422    2.6172
         0         0    4.1172    2.8125
         0         0         0    3.8359

=== 絕對誤差 ===
    0.0228    0.0088    0.0170    0.0115
         0    0.0309    0.0068    0.0010
         0         0    0.0050    0.0149
         0         0         0    0.0126

SQNR : 47.22 dB
Golden Pattern 產生成功！
fx >>
```

# III. Verilog verification

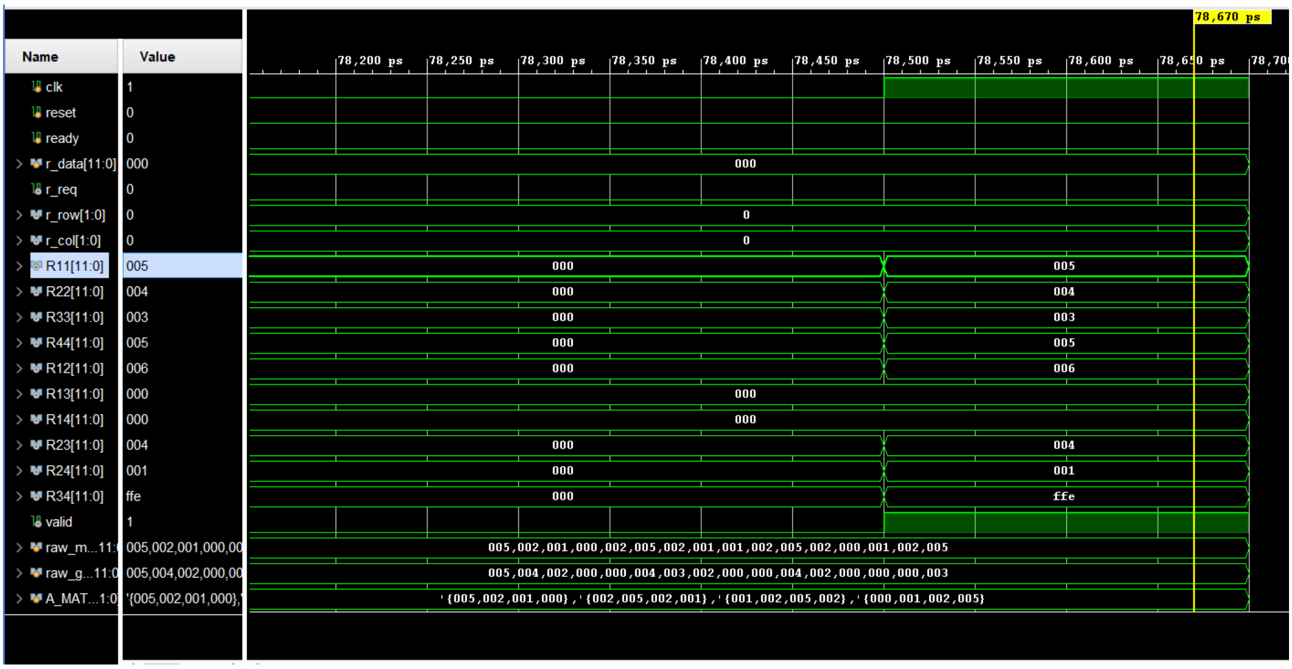
```
Tcl Console    Messages    Log

==================================================
   Input Matrix A (8-bit integers)
==================================================
  |    5     2     1     0 |
  |    2     5     2     1 |
  |    1     2     5     2 |
  |    0     1     2     5 |

==================================================
   Golden R Matrix (Q5.7 fixed-point)
==================================================
  |     5      4      2      0 |
  |     0      4      3      2 |
  |     0      0      4      2 |
  |     0      0      0      3 |

==================================================
   HW R Matrix (Q5.7 fixed-point)
==================================================
  |     5      6      0      0 |
  |     0      4      4      1 |
  |     0      0      3     -2 |
  |     0      0      0      5 |

==================================================
```


# IV. Verilog synthesis

Tcl Console | Messages | Log | Reports | **Design Runs** | DRC | Methodology | Power | Timing

| Name | Constraints | Status | WNS | TNS | WHS | THS | WBSS | TPWS | Total Power | Failed Routes | Methodology | RQA Score | QoR Suggestions |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **synth_1** (active) | **constrs_1** | **synth_design Complete!** | | | | | | | | | | | |
| impl_1 | constrs_1 | route_design Complete! | 1.007 | 0.000 | 0.040 | 0.000 | | 0.000 | 0.090 | 0 | 616 Warn | | |

| LUT | FF | BRAM | URAM | DSP | Start | Elapsed | Run Strategy | Report Strategy | Part |
|---|---|---|---|---|---|---|---|---|---|
| **1783** | **931** | **0** | **0** | **16** | **6/15/26, 3:08 PM** | **00:00:59** | **Vivado Synthesis Defaults (Vivado Synthesis 2024)** | **Vivado Synthesis Default Reports (Vivado Synthesis 2024)** | **xc7a35tcsg324-1** |
| 1754 | 1001 | 0 | 0 | 16 | 6/15/26, 3:13 PM | 00:01:08 | Vivado Implementation Defaults (Vivado Implementation 2024) | Vivado Implementation Default Reports (Vivado Implementation 2024) | xc7a35tcsg324-1 |

1. Total logic gates: LUT:1754, FF:1001, DSP:16

```
Report Cell Usage:
+------+--------+------+
|      |Cell    |Count |
+------+--------+------+
|1     |BUFG    |     1|
|2     |CARRY4  |   240|
|3     |DSP48E1 |    16|
|4     |LUT1    |    41|
|5     |LUT2    |   278|
|6     |LUT3    |   312|
|7     |LUT4    |   142|
|8     |LUT5    |   244|
|9     |LUT6    |   811|
|10    |FDCE    |   920|
|11    |FDPE    |    11|
|12    |IBUF    |     8|
|13    |OBUF    |   126|
+------+--------+------+
```

2. Maximum working frequency: 1000/(15-1.007) = 71.46MHZ

3. Clock cycles needed to complete QR: (78.5-3)/15 = 5 clock cycles

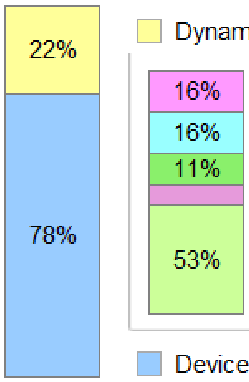
4. Throughput rate: ($71.46*10^6$ HZ) / (5 cycles) =14,292,000/sec

5. Power consumption at maximum working frequency:

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

| | |
|---|---|
| **Total On-Chip Power:** | **0.09 W** |
| **Design Power Budget:** | **Not Specified** |
| **Process:** | typical |
| **Power Budget Margin:** | **N/A** |
| **Junction Temperature:** | **25.4°C** |
| Thermal Margin: | 59.6°C (12.4 W) |
| Ambient Temperature: | 25.0 °C |

**On-Chip Power**




| | | |
|---|---|---|
| Dynamic: | 0.020 W | (22%) |
| Clocks: | 0.003 W | (16%) |
| Signals: | 0.003 W | (16%) |
| Logic: | 0.002 W | (11%) |
| DSP: | 0.001 W | (4%) |
| I/O: | 0.010 W | (53%) |
| Device Static: | 0.070 W | (78%) |

6. Energy needed to complete = 0.09 / 14292000 = 6.3nJ

7. timing report:

**Design Timing Summary**

| **Setup** | | **Hold** | | **Pulse Width** | |
|---|---|---|---|---|---|
| Worst Negative Slack (WNS): | 1.007 ns | Worst Hold Slack (WHS): | 0.040 ns | Worst Pulse Width Slack (WPWS): | 7.000 ns |
| Total Negative Slack (TNS): | 0.000 ns | Total Hold Slack (THS): | 0.000 ns | Total Pulse Width Negative Slack (TPWS): | 0.000 ns |
| Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 | Number of Failing Endpoints: | 0 |
| Total Number of Endpoints: | 3008 | Total Number of Endpoints: | 3008 | Total Number of Endpoints: | 1002 |

**All user specified timing constraints are met.**